FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 01-08-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABCD

Sol. MB(cos 0° – Cos 90°) = $\frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2$ …(1)

Also, $I_1\omega_1 = I_2\omega_2$ …(2)

By (1) and (2) we get

$K_{max} = \frac{\mu_0\pi r^2 i_1 i_2 MR}{2(MR^2 + mr^2)}$

2. BC

Sol. As $\lambda_0 = \frac{hc}{eV}$ and as V increases. λ_0 decreases but characteristic wavelengths do not change so interval between $\lambda_{k\alpha}$ & λ_0 increases and the same for the interval between $\lambda_{k\beta}$ & λ_0

3. ABD

Sol. By conservation of angular momentum,

$I_1\omega_i = (I_1 + I_2)\omega_f$

$\omega_f = \omega_1 / 3$

$E_i = \frac{1}{2}I_1\omega_i^2$

$E_f = \frac{1}{2}(I_1 + I_2)\omega_f^2$

Ratio of the heat produced to initial kinetic energy $= \frac{E_i - E_f}{E_i} = \frac{2}{3}$

4. AB

Sol. $-\frac{dv}{dt} = \frac{v^2}{R}$ & $-v\frac{dv}{ds} = \frac{v^2}{R}$

5. AB

Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

6. ABCD

Sol. $\vec{F} = 3\hat{i} - 4\hat{j}, W_{con.f} = U_i - U_f = 34 - 9 = 25$ & $\frac{1}{2}mv^2 - 16 = 34$

SECTION – B

7. 7

Sol. For equilibrium of bubble.

$4\pi R^2 d\,\rho_{sol} g + \frac{4}{3}\pi R^3\,\rho_{He} g = \frac{4}{3}\pi R^3 \rho_{air} g$

$d \rightarrow$ Thickness of soap solution.

$\therefore$ d = 3.5 μm.

8. 2

Sol. Use formula in terms of reduced mass

$$\frac{1}{\lambda} = \frac{R_\infty z^2}{\left(1 + \frac{m}{M}\right)}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\therefore \quad \frac{\lambda_D}{\lambda_H} = \frac{1 + \frac{m}{M_D}}{1 + \frac{m}{M_H}}$$

$$\frac{M_D}{M_H} \approx 2.0$$

9. 7

Sol. For process AB $T_A = 300\ K,\ T_B = 600\ K$

$W = nR\Delta T = nR(T_B - T_A) = 300\ nR = 600R.$

$Q = n\ Cp\ \Delta T = 2 \times \frac{5}{2} R\ (300) = 1500R.$

For process BC $W = nRT\ln\frac{v_f}{v_c} = nRT\ln\frac{p_i}{p_f} = nRT\ln 2 = 1200R\ln 2$

$Q = W = 1200R\ln 2$

For process CA $W = \int P\,dV = \int_{600}^{300}\frac{K}{T}\frac{2nRT}{K}dT.$

$= -2nR(300) = -1200R.$

$Q = nC_V\,\Delta T + W$

$= 2\times\frac{3}{2}R(-300) - 1200R.$

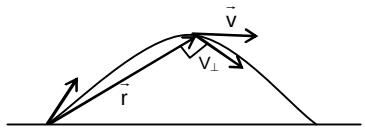
$= -900R - 1200R = -2100R$

$\eta = \frac{600R + 1200R\ln 2 - 1200R}{1500R + 1200R\ln 2}$

$= 1 - \frac{21}{12\ln 2 + 15} \Rightarrow x = 7.$

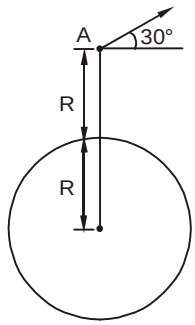
10. 2

Sol. $\omega = \frac{V_\perp}{|\vec{r}|}$



11. 3

Sol. Let v_0 be the velocity required to send a particle to infinity from point A.

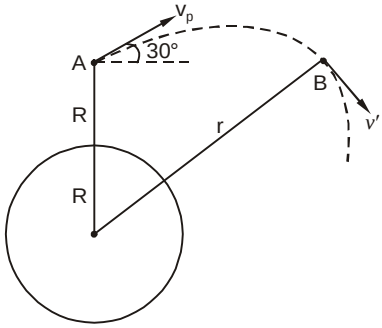


$\frac{1}{2}mv_0^2 - \frac{GMm}{2R} = 0$

$\Rightarrow \quad v_0 = \sqrt{\frac{GM}{R}}$

$\therefore$ velocity of projection at A

$v_p = \frac{1}{\sqrt{2}}v_0 = \sqrt{\frac{GM}{2R}}$



Let v' be the velocity at the highest point and r be its distance from the center of the plant A.

$\frac{1}{2}mv_0^2 - \frac{GMm}{2R} = 0 \quad \Rightarrow v_0\sqrt{\frac{GM}{R}}$

$\therefore$ velocity of projection at A

$v_p = \frac{1}{\sqrt{2}} v_0 = \sqrt{\frac{GM}{2R}}$

Let v' be the velocity at the highest point and r be its distance from the center of the planet. Using Conservation of angular momentum.

$m\, v_p\, 2R\cos 30° = m\, v' r$

$\Rightarrow \quad v' = \frac{\sqrt{3}R}{r}\sqrt{\frac{GM}{R}} = \sqrt{\frac{3GMR}{2r^2}} \qquad \ldots(1)$

From conservation on the mechanical energy at A and B

$-\frac{GMm}{2R} + \frac{1}{2}m.\frac{GM}{2R} = -\frac{GMm}{r} + \frac{1}{2}mv'^2$

replacing the value of v' from (1) in the above eq^n.

$-\frac{GM}{4R} = -\frac{GM}{r} + \frac{1}{2}\times\frac{3}{2}\frac{GMR}{r^2} \Rightarrow r^2 - 4Rr + 3R^2 = 0$

$\Rightarrow (r - 3R)(r - R) = 0 \qquad \Rightarrow r = R,\ r = 3R$

12. 2

Sol. Velocity of efflux = $16 \times 0.25 = 4$ m/s^2

Time of fall of the liquid = $\sqrt{\frac{2h}{g}} = 0.25$ sec.

Thus, range on horizontal surface = velocity of efflux × time of fall = 2 m.

SECTION – C

13. 00001.50

Sol. $i_1 = \frac{E}{\frac{r}{m} + \frac{R}{n}}$ & $i_2 = \frac{E}{\frac{r}{n} + \frac{R}{m}}$

$\frac{i_2}{i_1} = \frac{\frac{r}{m} + \frac{R}{n}}{\frac{r}{n} + \frac{R}{m}} = \frac{1 + \frac{m}{n}\cdot\frac{R}{r}}{\frac{m}{n} + \frac{R}{r}} = \frac{3}{2} = 1.5$

14. 00003.73

Sol. $V\left(x = \frac{A}{2}\right) = \sqrt{3}\,\omega\frac{A}{2}$

$T_1 = 2\times\left(\frac{T}{4} + \frac{T}{12}\right) = \frac{2T}{3} \qquad \left(\text{SHM Part} : T = \frac{2\pi}{\omega}\right)$

$T_2 = 2\times\frac{\sqrt{3}\,\omega\frac{A}{2}}{\frac{A\omega^2}{2}} = \frac{2\sqrt{3}}{\omega}$ (Constant acceleration part)

Time period $= T_1 + T_2$

$= \frac{4\pi}{300} + \frac{2\sqrt{3}}{\omega}$

15. 00629.20

Sol. $X_L = 50\ \Omega$, $X_C = 100\ \Omega$

$Z_1 = 100 - 50\ i$; $Z_2 = 100 + 100\ i$ $(i = \sqrt{-1})$

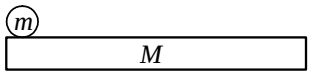
$$Z = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{100}{17}(13 + i)$$

$$|Z| = \frac{100}{17}\sqrt{170}\ ;\ \cos\theta = \frac{13}{\sqrt{170}}$$

$$P = \frac{V^2}{|Z|}.\cos\phi = 629.20$$

16. 00001.40

Sol. $V_{mM} = 4\hat{i} + 3\hat{j}$



$V_m = u\hat{i} + 3\hat{j}$

$V_M = -V\hat{i}$

Here, along horizontal direction $mV_m + MV_M = 0$

$$\Rightarrow \quad V = \frac{um}{M} = u\eta \text{ and } u + V = 4 \Rightarrow u = \frac{4}{1+\eta}$$

To move one meter w.r.t. ground

$$1 = \frac{2 \times \frac{4}{1+\eta} \times 3}{g} \Rightarrow 1 + \eta = \frac{24}{g} \text{ or } \eta = \frac{14}{10} = 1.40$$

17. 00252.98

Sol. Loudness (dB) = $10 \log_{10}\left(\frac{I}{I_o}\right)$ ($I_o \rightarrow$ Minimum audible intensity)

$$30 = 10 \log_{10}\left(\frac{I}{I_o}\right)$$

$$\frac{I}{I_o} = 1000$$

$$\Rightarrow \quad I = (1000)I_o$$

For point source, intensity $\propto \frac{1}{(\text{Distance})^2}$

$$(1000)I_o \propto \frac{1}{(8)^2}$$

$$\Rightarrow \quad I_o \propto \frac{1}{(8)^2\,(1000)}$$

$\Rightarrow$ Required distance = $8\sqrt{1000} = 80\sqrt{10}$

$\Rightarrow$ 252.98 m

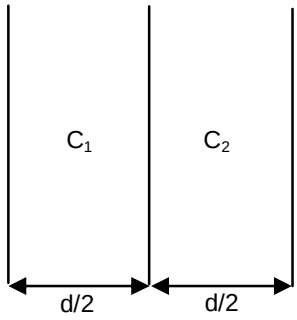
18. 00001.72

Sol. $$\frac{1}{C_1}=\frac{1}{A}\int_0^{d/2}\frac{dx}{\in_0+\beta x}$$

$$\frac{1}{C_2}=\frac{1}{A}\int_{d/2}^{d}\frac{dx}{\{\in_0+\beta(d-x)\}}$$

$$\frac{1}{c}=\frac{1}{c_1}+\frac{1}{c_2}$$

$$K_{eq}=\frac{\beta d}{2\in_0 \ell n\left(\frac{\in_0+\frac{\beta d}{2}}{\in_0}\right)}$$



Chemistry

PART – II

SECTION – A

19. ABCD

Sol. $A(g) \rightleftharpoons B(g) + C(g) + D(g)$

1-x x-y x+y x-y

$B(g)+D(g) \rightleftharpoons C(g) + E(g)$

x-y x – y x+y y

Total pressure = $P_A + P_B + P_C + P_D + P_E$

$(1-x)+(x-y)+(x+y)+(x-y)+y = 1+2x$

$\because P_T = 2P_A$

$2(1-x) = 1+2x$

$x = ¼$

$P_c=(x+y)=3/8$ (Given)

Therefore $y = 1/8$

$$K_{C_1} = \frac{P_B \times P_C \times P_D}{P_A} = \frac{\frac{1}{8}\times\frac{3}{8}\times\frac{1}{8}}{\frac{3}{4}} = \frac{1}{128}$$

$$K_{C_2} = 3\,, K_C = \frac{3}{128}$$

20. ACD

Sol. State function only depend on initial and final state

21. ABC

Sol. Net of neutralization with strong acid & base is constant, which is =–13.7 K cal/ gm equivalent.

22. ABC

Sol. $\bar{\nu} = \frac{1}{\lambda} = \left(\frac{1}{2^2} - \frac{1}{4^2}\right) RZ^2 = \frac{3}{4}R$

In H–spectrum for the same $\bar{\nu}$ or λ as Z = 1, n = 1, $n_2 = 2$

23. BCD

Sol. Bond order of $O_2^+ = \frac{1}{2}[8-3] = \frac{5}{2}$

Bond order of $N_2^+ = \frac{1}{2}[7-2] = \frac{5}{2}$

Bond order of $N_2^{2-} = \frac{1}{2}[8-4] = 2$

Bond order of $O_2^- = \frac{1}{2}[8-5] = \frac{3}{2}$

Bond order of $He_2 = \frac{1}{2}[2-2] = 0$

24. AB

Sol. When a solid melts there will be an increase in enthalpy & a decrease in free energy.

SECTION – B

25. 3

Sol. $2X(g) \rightleftharpoons 2Y(g) + Z(g)$

$1 - \alpha \qquad \alpha \qquad \frac{\alpha}{2}$ Total moles = $1 + \frac{\alpha}{2} = \frac{2+\alpha}{2}$

$$K_P = \frac{(P_Y)^2 (P_Z)}{(P_X)^2}$$

$$= \frac{\left(\frac{2\alpha}{(2+\alpha)}\right)^2 P^2 . \left(\frac{\alpha}{2+\alpha}\right) P}{\left(\frac{2(1-\alpha)}{(2+\alpha)}\right)^2 P^2}$$

neglecting α compared to 1 and solving

$$\alpha = \left(\frac{2K_P}{P}\right)^{1/3} \Rightarrow n = 3$$

26. 2

Sol. $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}} - CH_3$

27. 7

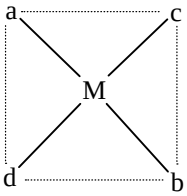
Sol. $2KMnO_4 + KI + H_2O \rightarrow 2KOH + 2MnO_2 + KIO_3$

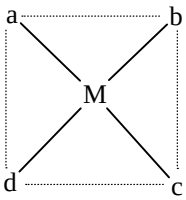
3.5 moles of potassium iodide requires 7 mol of potassium permanganate.

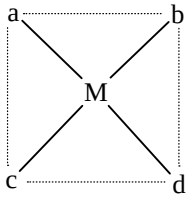
28. 3

Sol. $\left[Pt(NH_3)(NH_2OH)(NO_2)(py)\right]NO_2$

All four ligands are different, so it can be compared to M abcd. Three geometrical isomers are possible as follows







29. 5

Sol. Chiral C is attached to 4 different groups.

30. 1

Sol. Conc. H_2SO_4 is a dehydrating agent.

SECTION – C

31. 00022.40

Sol. Molar mass of the salt is 322

Moles of salt = $\frac{32.2}{322} = 0.1$

$\therefore$ Moles of oxygen = $14 \times 0.1 = 1.4$

Mass of oxygen = $1.4 \times 16 = 22.4$ g

32. 00231.60

Sol. $\Delta G^o = -nFE^o_{Cell}$

$= -3 \times 96500 \times (-0.8)$

$= 231600 \times 10^{-3} = 231.6$ kJ mol^{-1}

33. 00000.02

Sol. $\underset{(1-x)}{MgSO_4(s)} + aq \rightleftharpoons \underset{x}{Mg^{2+}(aq)} + \underset{x}{SO_4^{2-}(aq)}$

$K_{sp} = \left[Mg^{2+}\right]\left[SO_4^{2-}\right]$

$K_{sp} = x^2$

$x = \sqrt{K_{sp}} = \sqrt{4 \times 10^{-4}} = 0.02$

34. 00190.50

Sol. Reaction product of borazine with HCl is

$B_3N_3H_6 + 3HCl \longrightarrow \underset{(X)}{B_3N_3H_9Cl_3}$

35. 00000.04

Sol. Using concept of titration

$M_{eq}(NH_4)_2FeSO_4 = m_{eq}K_2Cr_2O_7$

$0.3 \times V \times 1 = 20 \times 6 \times 0.1$

$V = 0.04$

36. 00001.85

Sol. $I.E = \frac{R_H \times Z^{*2}}{n^2}$

Where Z^* = Effective nuclear charge

Mathematics

PART – III

SECTION – A

37. AB

Sol. $9\cos^{12} x + \cos^2 2x + 1 - 6\cos^6 x \cos 2x - 6\cos^6 x + 2\cos 2x = 0$

$\Rightarrow \quad \left(3\cos^6 x - 1 - \cos 2x\right)^2 = 0$

$\Rightarrow \quad \cos^2 x\left(3\cos^4 x - 2\right) = 0$

$\Rightarrow \quad \cos x = 0$

$x = n\pi + \frac{\pi}{2},\ n \in I$

or $\quad \cos^4 x = \frac{2}{3}$

$\cos x = \pm\sqrt[4]{\frac{2}{3}}$

$x = m\pi \pm \cos^{-1}\sqrt[4]{\frac{2}{3}},\ m \in I$

38. ABD

Sol. Let any point on the line through P be $x = \alpha + r\cos\theta,\ y = 2 + r\sin\theta$.

Put this in $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and $xy = 0$.

$PA \times PD = PB \times PC$

$\Rightarrow\ 2\alpha\sin 2\theta + 5\cos 2\theta = 13$

$\Rightarrow\ 13 \le \sqrt{4\alpha^2 + 25}$

$\Rightarrow\ |\alpha| \ge 6$

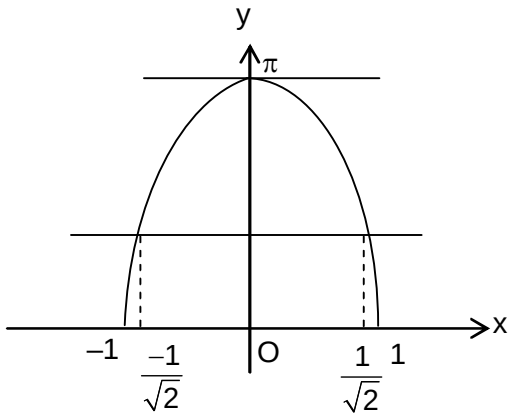
39. AC

Sol. Differential coefficient of $f(x)$ w.r.t.

$\sin^{-1} x = -2\frac{x}{|x|}$ at $x = \frac{1}{2}$ it is –2

$f(0^+) = -2; f'(0^-) = 2$

$\Rightarrow$ f is not derivable at x = 0

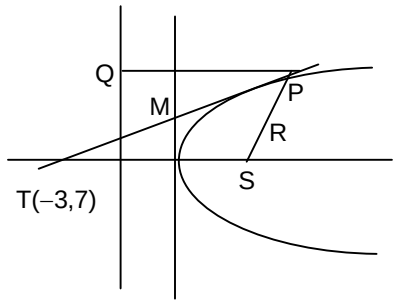


40. ABC

Sol. $f(x) = \frac{\sin x}{x}$

41. ABCD

Sol. Equation of tangent at P is x+y=4



Clearly mirror image of R(–5,5) lies on line PQ.

Now mirror image R′ or R

$\Rightarrow \frac{\alpha'+5}{1}=\frac{\beta'-5}{1}=\frac{-2(-5+5-4)}{2}=4$

$\Rightarrow (\alpha',\beta')\equiv(-1,9)$

Let PM cuts the axis at T; As M is midpoint of PT

$\Rightarrow$ T is (–3, 7)

We know that SP = ST and $\angle$SMP= $\frac{\pi}{2}$

Equation SP $\equiv y-3=-\frac{1}{3}(x-1)$

$\Rightarrow x + 3y–10 = 0$

Let $S \equiv (10 - 3\beta, \beta)$

Again TS || PQ $\Rightarrow \frac{\beta-7}{13-3\beta}=\frac{9-3}{-1-1}=-3$

$\therefore$ focus is (–2, 4)

42. AD

Sol. $h'(x) = f(x) \Rightarrow h''(x) = f'(x)$

$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = g(1)$

$f(g(x)) = x$

$f'(g(x)).\ g'(x) = 1$

$f'(g(1)).\ g'(1) = 1 \Rightarrow g'(1) = 1$

$G(x) = x^2.g(x) – x.\ h(g(x))$

$G'(x) = x^2\ g'(x) + 2x.\ g(x) – h(g(x)) – x.\ h'(g(x)).\ g'(x)$

$= 2x.g(x) – h(g(x))$

$G''(x) = 2.g(x) + x.g'(x)$

$G'(1) = 2g(1) – h\ (g(1)) = 2g(1) – h(1) = 2$

$G''(1) = 2.g(1) + g'(1) = 3$

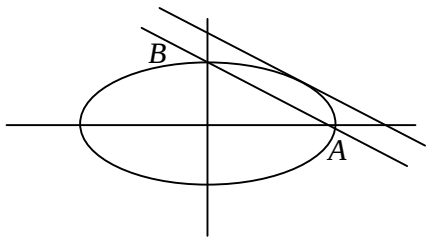
SECTION – B

43. 3

Sol. $XX^T = I$; Find PQ then $(PQ)^{10}$

44. 3

Sol. Let perpendicular distance of *P* from the line be *h*



$\frac{1}{2}\times h\times 5=6(\sqrt{2}-1)$ (as $\Delta PAB=6(\sqrt{2}-1)$)

$\Rightarrow h=\frac{12(\sqrt{2}-1)}{5}$

Now distance of tangent parallel to *AB* i.e. $4y+3x=12\sqrt{2}$, from line *AB* is $\frac{12(\sqrt{2}-1)}{5}$.

There are just three such points.

45. 5

Sol. Note that $z = 0$ is one root of the given equation

$z^4 + az^3 + (12 + 9i)z^2 + bz = 0$

canceling z, we get $z^3 + az^2 + (12 + 9i)z + b = 0$ …(1)

The 3 vertices of the square other than $(0 + 0i)$ are

u, iu, u (1 + i)

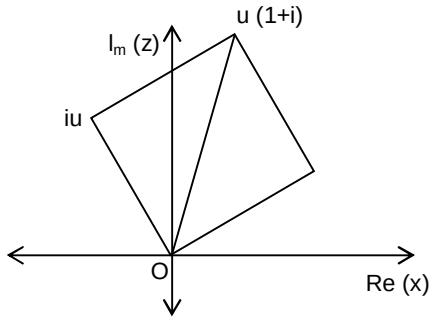
Now $u(ui) + (iu) + (1+i)u + [(1+i)u]u = 12 + 9i$

(using theory of equation)

$u^2 = \dfrac{4+3i}{i} = 3 - 4i \Rightarrow u = \sqrt{3-4i} = \pm(2 - i)$

$\therefore u = 2 - i$ or $-2 + i$

$\Rightarrow |u|^2 = 5$



46. 3

Sol. $\beta = \displaystyle\int_0^1 \frac{2x + 3x^2 + 4x^3}{2\sqrt{x^4 + x^3 + x^2}}\,dx$

Put $(x^2 + x^3 + x^4) = t \quad \Rightarrow \quad (2x + 3x^2 + 4x^3)\,dx = dt$

$\beta = \displaystyle\int_0^3 \frac{dt}{2\sqrt{t}}$

$\beta = \left[\sqrt{t}\right]_0^3 = \sqrt{3} \quad \Rightarrow \quad \beta^2 = 3$

47. 6

Sol. $\dfrac{dy}{dt} = -k\sqrt{y}$

$\dfrac{dy}{\sqrt{y}} - k\,dt$

$2\sqrt{y} = -kt + c$

$t = 0;\ y = 4$

$4 = c$

$2\sqrt{y} = -kt + 4$

$y = 0$

$kt = 4$

$t = \dfrac{4}{\frac{1}{15}} = 60$ minutes

48. 3

Sol. $y = mx$ touches circle

$\Rightarrow D = 0$ for $(m^2 + 1)x^2 - 2a(m + 2)x + 4a^2 = 0$

$\Rightarrow 4a^2(m + 2)^2 = 4(m^2 + 1)(4a^2)$

$\Rightarrow m = 0, \frac{4}{3}$

Let $A:(x_1, m_1x_1), B:(x_2, m_2x_2)$

$AB^2 = 9 \Rightarrow (x_1 - x_2)^2 + (m_1x_1 - m_2x_2)^2 = 9$ where

$x_1 = \frac{(m_1 + 2)}{m_1^2 + 1}a,\ x_2 = \frac{(m_2 + 2)}{m_2^2 + 1}a,\ m_1 = 0,\ m_2 = \frac{4}{3}$

get $a = \frac{3\sqrt{5}}{4}$

SECTION – C

49. 00000.30

Sol. Let $S = \sum_{k=0}^{100}\left(\frac{k}{k+1}\right)^{100}C_k$

$= \sum_{k=0}^{100}\frac{((k+1)-1)}{(k+1)}{}^{100}C_k = \left(\sum_{k=0}^{100}{}^{100}C_k\right) - \sum_{k=0}^{100}\frac{{}^{100}C_k}{k+1}$

$= 2^{100} - \frac{1}{101}\sum_{k=0}^{100}\left(\frac{101}{k+1}{}^{100}C_k\right)$

$= 2^{100} - \frac{1}{101}\sum_{k=0}^{100}{}^{101}C_{k+1}$

$2^{100} - \left(\frac{2^{101}-1}{101}\right) = \frac{(101)2^{100} - 2^{101} + 1}{101} = \frac{99\left(2^{100}\right)+1}{101}$

$= \frac{a\left(2^{100}\right)+b}{c}$ (Given)

So, $a = 99, b = 1, c = 101$

Hence, $(a+b+c)_{least} = 99 + 1 + 101 = 201.$

50. 00000.14

Sol. Let $I = \lim_{x\to 1}\left(\frac{9}{1-x^9} - \frac{7}{1-x^7}\right)$ ……(i)

Put $x = \frac{1}{y}$, we get $I = \lim_{x\to 1}\left(\frac{-9x^9}{1-x^9} + \frac{7x^7}{1-x^7}\right)$ ……(ii)

$\therefore$ On adding equation (i) and (ii), we get

$2I = 9 - 7 = 2$

$\Rightarrow I = 1$

Common ratio $= \lim_{x\to 0}\frac{\tan x - \sin x}{\ln^3(1+x)}$

$= \lim_{x\to 0}\frac{\tan x - \sin x}{x^3\left(\frac{\ln(1+x)}{x}\right)^3}$

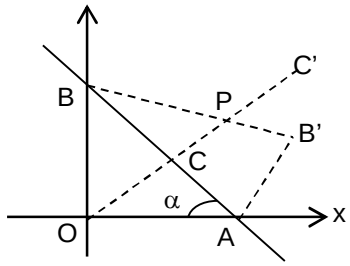
$$= \lim_{x\to 0} \frac{\tan x\left(1-\cos x\right)}{x^2} = \frac{1}{2}$$

51. 00002.00

Sol. $A \equiv (1,0)$, $B \equiv \equiv (0, \tan\alpha)$, $C \equiv \left(\frac{1}{2}, \frac{\tan\alpha}{2}\right)$,

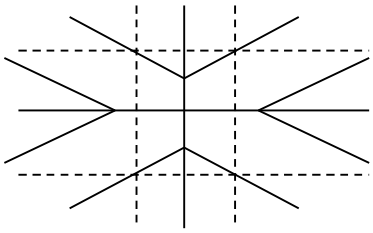
$B' = \left(\frac{2\tan^2\alpha}{1+\tan^2\alpha}, \frac{\tan^3\alpha - \tan\alpha}{1+\tan^2\alpha}\right)$

$\frac{\Delta ABB'}{\Delta BB'C} = 2$



52. 00002.00

Sol.



$x = \frac{-1}{2}$ $\quad x = \frac{1}{2}$

53. 00005.00

Sol. Let $\vec{d}.\vec{a} = \cos y\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = -\vec{d}.\left(\vec{b}+\vec{c}\right)$ [as $\vec{d}.\left(\vec{a}+\vec{b}+\vec{c}\right) = 0$]

$$\Rightarrow \quad \cos y = -\frac{\vec{d}.\left(\vec{b}+\vec{c}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]} \qquad \text{.....(1)}$$

similarly $\sin x = -\frac{\vec{d}.\left(\vec{b}+\vec{a}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ .....(2)

$$\Rightarrow 2 = -\frac{\vec{d}.\left(\vec{a}+\vec{c}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]} \qquad \text{.....(3)}$$

Adding these we get $\sin x + \cos y + 2 = 0$

$\Rightarrow \sin x + \cos y = -2$

$\Rightarrow \sin x = -1, \cos y = -1$

$\Rightarrow x = (4n-1)\frac{\pi}{2},\ y = (2n+1)\pi$

Since we want minimum value of $x^2 + y^2$, so $x = -\frac{\pi}{2},\ y = \pi$

$\Rightarrow x^2 + y^2 = \frac{5\pi^2}{4} \Rightarrow \lambda = 5$

54. 00005.00

Sol. Equation of normal at point $T\left(t,\frac{t^2}{2}\right)$ to C is $y-\frac{t^2}{2}=\frac{-1}{t}(x-t)\Rightarrow y=\frac{-x}{t}+\frac{t^2}{2}+1$

If circle S touches curve C at T.

$\Rightarrow (x(t),y(t))$ must lie on normal

$\Rightarrow y(t)=\frac{-x(t)}{t}+\frac{t^2}{2}+1$

Also distance of centre of circle from T equals radius = distance of centre of circle from x – axis.

$\Rightarrow (x(t)-t)^2+\left(y(t)-\frac{t^2}{2}\right)^2=(y(t))^2$

$x(t)=\frac{t\left(1+\sqrt{t^2+1}\right)}{2},\ y(t)=\frac{t^2+1-\sqrt{t^2+1}}{2}$

$\Rightarrow x(t)y(t)=\frac{1}{4}t^3\sqrt{t^2+1}$

$\lim\limits_{r\to 0^+}\int\limits_r^1\frac{1}{4}t^3\sqrt{t^2+1}\,dt$

Let $\sqrt{1+t^2}=z\Rightarrow 2tdt=2z\,dz$

$\Rightarrow \lim\limits_{r\to 0^+}\int\limits_{\sqrt{1+r^2}}^{\sqrt{2}}\frac{1}{4}(z^2-1)z^2dz=\frac{\sqrt{2}+1}{30}$